ASCOMP 2026 POSTER

COOPERATIVE GAMES / FEATURE SELECTION

Min-Marginal Power Indices for Multi-Objective Feature Selection

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Find when a feature can hurt.
Verify removal before filtering.

12.53%
mean analyzed-feature removal

-0.000313
mean held-out utility change

01 THE PROBLEM

Averages can hide a harmful context.

Adding a feature can turn an acceptable model into an unacceptable one in a particular coalition.

ONE COALITION, ONE HARMFUL WITNESS

Acceptable
$v(S) = 1$

+ feature i
$\Delta = -1$

Unacceptable
$v(S \cup \{i\}) = 0$

A worst-case marginal

$$v_\alpha(S) = 1[U(M_S) \geq \alpha\, U_{full}]$$

$$\Delta_i(S) = v_\alpha(S \cup \{i\}) - v_\alpha(S)$$

$$\mathbf{MinPI_i = \min \Delta_i(S)}$$

over all $S \subseteq N \setminus \{i\}$; N is the analyzed feature set

-1 harmful	0 neutral*	+1 decisive

*Exact values. A sampled 0 or +1 is not a certificate; sampling can miss witnesses.

Scan to read the full paper

Theory, experiments and complete results

hlnks.co/63671101

02 THE METHOD

1 Discover candidates

Fit coalition models on training data.
Evaluate threshold crossings on validation data.

Exact enumeration or sampled search

2 Gate proposed removals

Let H be the candidate harmful set.
Accept its removal only if:

$$U_{val}(N \setminus H) \geq U_{val}(N) - \varepsilon$$

ε = 0.005 (classification); 0.01 (regression)

PASS: remove H | FAIL: remove nothing

3 Evaluate & optimize

Use held-out test data for final reporting.
Accepted subsets may prefilter NSGA-II.

EXPERIMENTAL PROTOCOL

6 datasets × 5 seeds

64% train / 16% validation / 20% test

XGBoost; α = 0.95; at most 100 analyzed features after training-only preprocessing.

ROC AUC, OVR AUC or R^2.
Controls delete the same accepted count.

03 THE EVIDENCE

Compression is dataset-dependent.

Five-seed mean ± SD. Bars: analyzed features removed.
ΔU: held-out utility minus the full model.

	COMPRESSION (%)	HELD-OUT ΔU
Madelon	43.60 ± 31.96	**+0.00469** ± 0.00497
Abalone	27.50 ± 25.62	**-0.00565** ± 0.01077
Credit-Card-Fraud	2.76 ± 2.89	**-0.00078** ± 0.00335
Spambase	0.72 ± 0.99	**-0.00003** ± 0.00007
Gisette	0.60 ± 0.89	**-0.00011** ± 0.00016
ISOLET	0.00 ± 0.00	**0.00000** ± 0.00000

0 40 80

14/30 units had nonzero removal; 3/17 nonempty candidate sets were vetoed by the gate.

NSGA-II: equal search budgets

+0.01584
mean held-out ΔHV
800 evaluations per variant

Holm p = 0.4375; not significant. Madelon drives most of the gain.

WHAT THIS DOES NOT ESTABLISH

No overall superiority or speedup. The validation gate does not guarantee held-out non-inferiority.